

FEDERICO BRUZZONE ✧ CURRICULUM VITAE

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Presently a PhD Candidate in Computer Science. I have a strong passion for **Programming Languages** and **Compilers**, and I'm currently diving into **AI (ML/DL) compilers**. For more information, just visit my personal [webpage](#).

✧ SCIENTIFIC PUBLICATIONS ✧

Journal Ranked Q1 on Scimago — — [bib](#) [pdf](#) [IEEE](#) [arXiv](#)

Journal Ranked Q1 on Scimago — — [bib](#) [pdf](#) [SpringerLink](#) [arXiv](#)

PREPRINTS

- [arXiv](#)
- [arXiv](#)
- [arXiv](#)
- [arXiv](#)
- [arXiv](#)

✧ EDUCATION ✧

- 2024–Present** PhD Candidate in Computer Science at the ADAPT Lab, University of Milan
- 2022–2024** MSc in Computer Science at University of Milan (*110/110 cum laude*, W. Cazzola, 15/07/2024)
Thesis: Toward a Modular Approach for Type Systems and LSP Generation
- 2019–2022** BSc in Musical Computer Science at University of Milan (13/10/2022)
- 2011–2019** Piano and Music Composition at I.S.S.M. Novara Conservatory
- 2014–2019** Diploma in Computer Science and Telecommunications at E. Alessandrini

✧ OPEN SOURCE CONTRIBUTIONS ✧

- 2026** IREE: [\[Codegen\]\[LLVMCPU\]](#) Hoist in-loop stack alloca
- 2026** IREE: [\[LLVMCPU\]](#) Fix scalable vectorization fallback on SME-only (no +sve) targets
- 2026** MLIR/LLVM: [\[mlir\]\[sparse\]](#) Avoid vectorizing non-contiguous COO coordinate loads
- 2026** MLIR/LLVM: [\[mlir\]\[ArmSVE\]](#) Fix comment inconsistency for `pack_lhs` in `ArmSVE/pack-unpack-mmt4d.mlir` (NFC)
- 2026** MLIR/LLVM: [\[mlir\]\[vector\]](#) add consistent stride verification to masked load/store and gather/scatter ops
- 2026** MLIR/LLVM: [\[mlir\]\[ArmSME\]](#) fix f64 scalable matmul crashes in `VectorLegalizationPass`
- 2026** IREE: [\[LLVMCPU\]](#) Fix ArmSME requiring classic SVE, breaking SME-only targets (e.g., Apple Silicon)
- 2026** IREE: [\[LLVMCPU\]](#) Add SME lowering-strategy tests for f64 and unsupported i8 matmuls
- 2026** MLIR/LLVM: [\[mlir\]\[VectorToLLVM\]](#) add opt-in `enable-gep-inbounds-nuw` pass flag for `vector.load/store`
- 2026** MLIR/LLVM: [\[mlir\]\[MemRefToLLVM\]](#) fix incorrect `nw` on GEP/mul when lowering `memref.load/store` with negative strides
- 2026** MLIR/LLVM: [\[NFC\]\[mlir\]\[linalg\]](#) add `toContractionDimensions` for healthy code reuse
- 2026** MLIR/LLVM: [\[mlir\]\[vector\]](#) extend `createReadOrMaskedRead/createWriteOrMaskedWrite` with permutation map support
- 2026** MLIR/LLVM: [\[mlir\]\[affine\]](#) emit `in_bounds` on `transfer_read/write` when statically provable in `affine-super-vectorize`

- From 2026** [The Eter Programming Language](#), a new programming language leveraging the MLIR/IREE ecosystem with tensors support.
- From 2026** Maintainer of the [LLVM Passview](#), a framework to run an incremental, reproducible study of LLVM optimization passes.
- From 2026** Maintainer of the [LLVM Pass Template](#), a C++ template project to quickly create new LLVM passes.
- From 2025** Maintainer of the [Papers on Compiler Optimizations: Analysis and Transformations](#), a curated list of scientific publications.
- From 2025** Maintainer of [scribe](#), a minimalist, opinionated LaTeX document class for technical writing and presentations.
- From 2025** Maintainer of the [Tide Compiler](#), a backend-agnostic IR and compiler framework written in **Rust**.
- 2025** [Generalizing Closeness centrality to weighted networks using Newman method](#) (issue #1384)
- 2025** Rust: [Use the type-level constant value `ty::Value` where needed](#)
- 2025** Rust: [Add TooGeneric variant to LayoutError and emit Unknown](#)
- 2024** Rust: [Report the note when specified in diagnostic::on_unimplemented](#)
- From 2024** Maintainer of the cross-platform [tgt](#) project and [tdlib-rs](#) TDLib bindings written in **Rust**.
- From 2024** [CHIP-8 STM32-Port](#): CHIP-8 and S-CHIP emulator ported to STM32 Cortex M4 microcontroller.
- From 2023** [CHIP-8 Core](#): Core CHIP-8 and S-CHIP emulator library, used as the foundation for platform ports.

◆ RESEARCH ACTIVITIES ◆

- May 2026** **Committee Member** at the Int. Conf. on Software Language Engineering ([SLE 2026](#)), *ACM*, **B on CORE**
- Dec 2025** **Reviewer** for the Eur. Conf. on Object Oriented Programming ([ECOOP 2026](#)), *ACM*, **A on CORE**
- Nov 2025** **Reviewer** for the Journal of Computer Languages ([COLA](#)), *Elsevier*, **C on CORE**
- Jun 2025** **Reviewer** for the Journal of Software and Systems Modeling ([SoSyM](#)), *Springer*, **Q1 on Scimago**
- 2–6 Jun 2025** **Participant** at [<Programming>2025](#) conference
- Apr–Jun 2025** **Committee Member** at the Int. Conf. on Software Language Engineering ([SLE 2025](#)), *ACM*, **B on CORE**
- Mar 2025** **Reviewer** for the Journal of Systems and Software ([JSS](#)), *Elsevier*, **Q1 on Scimago**
- Sep 2024** **Student Volunteer** at the Int. Conf. on Functional Programming ([ICFP 2024](#))

◆ DISSEMINATION ACTIVITIES ◆

- Mar 2026** [MLIR: Scaling Compiler Infrastructure for Domain Specific Computation](#) — Presentation of the paper by Lattner et al., 2021.
- Feb 2026** [Matheuristic Variants of DSATUR for the Vertex Coloring Problem](#) — Presentation of the paper by Dupin, 2024.
- Dec 2025** [Your Optimizing Compiler is Not Optimizing Enough. To Hell With Multiple Recursions!](#) — First MuseMI meeting of the 2025/2026 season, Milan.
- Jul 2024** [P4 Compiler in SDN](#) — Presentation on the P4 compiler in Software Defined Networking.
- Jul 2024** [PhD Project](#) — *Universal Language Server Protocol and Debugger Adapter Protocol for Modular Language Workbenches*, University of Milan.
- Jul 2024** [Master's Thesis](#) — *Toward a Modular Approach for Type Systems and LSP Generation*, University of Milan.

◆ TEACHING ACTIVITIES ◆

THESIS SUPERVISION

- 14/07/2026** M. Visconti, *Towards Neverlang3: A Generative Rust Approach for Resilient LL Parser Generators*, MSc, **102**
- 23/02/2026** D. Cerato, *Employing Metaprogramming: A Macro-Driven Strategy to Overcome Compiler-Imposed Limitations on Specialization*, BSc, **103**
- 10/04/2025** D. Pellegrino, *Scalable Multi-client Real-time Whisper*, BSc, **96**
- 09/04/2025** A. Longoni, *GUIDE: Graphical User Interface Development Environment*, MSc, **110L**
- 09/04/2025** L. Albani, *New Generalized Protocol For Software Product Line Extraction And Configuration*, MSc, **110L**

09/04/2025 G. Esposito, *Fr3D: A Framework for DAP-compatible DSL-oriented Debugging*, MSc, **110L**
 24/02/2025 L. Favini, *RustyEx: Instrumenting rustc to Extract Feature Dependency Graphs*, BSc, **102**

GRADUATE COURSES

2025–2026 Mathematical Logic (Art. 45), **BSc** in CS, University of Milan, S. Aguzzoli
 2024–2025 Programming 1 (Art. 45), **BSc** in CS, University of Milan, L. Capra (coordinator W. Cazzola)
 2024–2025 Mathematical Logic (Art. 45), **BSc** in CS, University of Milan, S. Aguzzoli
 2023–2024 Mathematical Logic, **BSc** in CS, University of Milan, S. Aguzzoli
 2023–2024 Computer Science, **BSc** in Comm. and Society, University of Milan, A. Momigliano
 2023–2024 Programming 1, **BSc** in CS, University of Milan, A. Trentini (coordinator P. Boldi)
 2023–2024 Programming in Python, **MSc** in Chemistry, University of Milan, M. Monga

ADDITIONAL ACTIVITIES

2023–Present **Private Tutoring** in Computer Science, Mathematics and Physics
 Nov 2024 **Collaborator** at the Bebras Challenge, ALaDDIn Lab
 Jan–Jun 2024 **Organizer** of the BSc Computer Science Laboratories, ALaDDIn Lab
 Jan–Jun 2024 **Collaborator** of the Workshops for schools, ALaDDIn Lab
 Nov 2023 **Collaborator** at the Bebras Challenge, ALaDDIn Lab

TECHNICAL SKILLS

Languages Rust, C/C++, Python, Go, OCaml, Java, Scala, Kotlin, Erlang, Lua, Dart, PHP, HTML/CSS, SQL, Bash, TeX
Systems/Tooling Git, CI/CD, Docker, GDB, Valgrind, Build Systems (e.g., CMake, Cargo, Pip), Cross-language Linking, Static/Dynamic Libraries, and FFI (e.g., C bindgen)
Area of Expertise Compiler Construction and Optimizations, Programming Languages Design, IR Design and Implementation, Type Systems, Language Support Tools (e.g., LSP), Static Analysis, Rustc Internals, MLIR, LLVM, Parsing Techniques and Parser Generators (e.g., ANTLR)

MUSICAL ACTIVITIES

2021–Present **Sound Engineer and Music Producer**, in collaboration with [Believe](#) and as a [SLAE](#) member.
 2006–Present **Pianist and Music Composer**, composing original pieces and arrangements for piano and various ensembles
 2019–Present **Piano and Music Teacher**, teaching piano and music theory and composition to students of all ages and levels

GRANTS AND FELLOWSHIPS

2024 [56th Top Github Public Contributor](#) in Italy out of 958
 2023–2024 Scholarship for the MSc in Computer Science, awarded by the University of Milan
 2020–2024 Scholarship for the BSc in Musical Computer Science, awarded by the University of Milan

TONGUES

Italian Mother tongue
English Level CEFR B2 (SLAM at University of Milan)
Spanish Base (A1–A2)

Milan, July 24, 2026